

VeritasAI: Advanced Fake News Detection

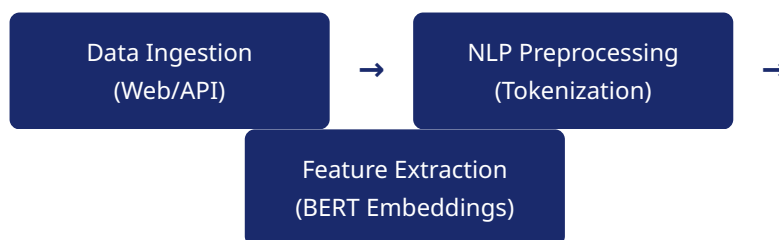
Idea Pitching Competition - Phase 1 Abstract

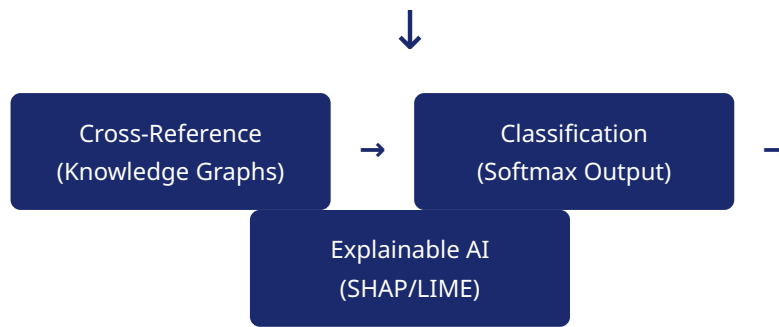
1. Project Abstract

*In an era of rapid digital information exchange, the proliferation of misinformation poses a significant threat to social stability and democratic integrity. **VeritasAI** is a hybrid deep learning framework designed to detect and categorize fake news with high precision. By combining **Transformer-based architectures (BERT/RoBERTa)** for semantic understanding with **Graph Neural Networks (GNNs)** to analyze propagation patterns, our solution goes beyond simple text analysis. We propose a multi-modal approach that validates textual claims against trusted knowledge graphs in real-time, providing users with a "Credibility Score" to restore trust in digital journalism.*

2. Methodology & Architecture

Our system operates through a specialized pipeline designed for scalability and low-latency inference:





Key Technical Innovations:

- **Contextual Awareness:** Utilizing Attention mechanisms to identify linguistic cues typical of clickbait and propaganda.
- **Fact-Checking Integration:** RAG (Retrieval-Augmented Generation) pipeline to verify claims against verified news repositories.
- **Explainability:** Using SHAP values to highlight exactly which words or phrases triggered a "Fake" flag, ensuring transparency for the end-user.

3. Competitive Edge

Unlike standard LSTM or CNN-based classifiers, VeritasAI addresses the *contextual evolution* of fake news. Our model is trained on diverse datasets including LIAR and FEVER, ensuring robustness against political, medical, and financial misinformation.

Tech Stack:

PyTorch HuggingFace LangChain FastAPI React.js Neo4j